

GAM: A Generative Auto-Marketing Framework in Online E-commerce Platforms

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Abstract

Auto-bidding plays an essential role in online advertising, allowing agents to automatically adjust bids for advertisers. Recently, the rise of Marketing Management service in e-commerce platforms has driven the evolution from auto-bidding to auto-marketing, enabling merchants to delegate their advertising bidding and product's coupon discounting decisions to agents. Auto-marketing requires agents to jointly decide on bidding and coupon discounting. Furthermore, compared to classic static constraints, auto-marketing agent faces a self-funding constraint (where the budget for both bidding and coupon discounting is entirely derived from the agent's commission revenue). Existing rule-based or RL-based methods often struggle with dynamic environments and complex sequential dependencies. To overcome these limitations, we propose a Generative Auto-Marketing framework (GAM), designed for performing joint sequential decisions on bidding and coupon discounting, and optimizing business objectives through post-training alignment. Furthermore, GAM employs a flexible, constraint-aware reward alignment module, and utilizes Group Relative Policy Optimization (GRPO) to align the pre-trained model, thus empirically balancing objective maximization and constraint satisfaction. We construct

an offline simulation environment based on large-scale real-world dataset, and demonstrate the effectiveness of GAM through extensive experimental results.

CCS Concepts

• **Information systems** → **Computational advertising**; • **Applied computing** → *Marketing*; *Online auctions*.

Keywords

Generative Model, Preference Alignment, Auto-marketing

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1 Introduction

In recent years, online e-commerce platforms have experienced rapid growth[11]. With the increasing multi-channelization and personalization of marketing services, coordinating marketing strategies across channels (e.g., online advertising and coupon discounting) has become a key challenge for merchants, promoting the demand for automated marketing services. Auto-bidding, a classic automated marketing service, allows agents to automatically adjust ad bids on behalf of advertisers, greatly simplifying the advertising process [1, 2]. However, auto-bidding only focuses on bidding and fails to include coupon discounting[33], which is another essential marketing channel. This consequently requires merchants to spend effort coordinating bidding and coupon discounting strategies, inevitably leading to a reduction in operational efficiency [15].

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Recently, Marketing Management service has emerged in major e-commerce platforms to further improve marketing efficiency, driving the evolution from auto-bidding to auto-marketing. Marketing Management service has been deployed on major e-commerce platforms, managing marketing strategies for hundreds of thousands of products daily while generating millions of transactions. Auto-marketing allows merchants to delegate their advertising bidding and product's coupon discounting decisions to agents, reducing the operational burden caused by separate configuration. In previous automated marketing services like auto-bidding, although agents could adaptively adjust bidding strategies, they still require merchants to pre-set static budgets or Key Performance Indicator (KPI) constraints to control risks [2, 16]. This leads to two main issues: first, setting appropriate static constraints for each channel is itself a complex task; second, static constraints lack flexibility, limiting the agent's capacity to adjust strategies based on real-time situations.

To address these issues, we propose a self-funding constraint in auto-marketing scenario, which better aligns with the essence of automated marketing services. Self-funding constraint requires that the agent's marketing costs (for bidding and coupon discounting) must be covered by commissions from product transactions, rather than obtaining pre-allocated budgets or KPI constraints from the merchant. Specifically, during the auto-marketing service, the agent automatically adjusts bids and coupon discounts on behalf of the merchant, and pays the corresponding costs. Simultaneously, auto-marketing agent collects a commission (usually a fixed percentage of the product's price) from the merchant per transaction to cover the marketing expenditure. The introduction of self-funding constraint shifts the burden of setting budgets and constraints from merchants to agents, effectively freeing merchants from complex marketing operations and granting agents greater decision-making autonomy.

Nevertheless, auto-marketing faces several inherent challenges. First, auto-marketing requires the agent to jointly decide on bidding and coupon discounting to maximize the cumulative value of won impressions. However, the decisions for ad bidding and coupon discounting are highly coupled, jointly impacting marketing performance. Second, the self-funding constraint requires the agent to balance costs and commissions throughout marketing period. Violating this constraint would result in platform deficits. Commonly used rule-based [9, 44] or Reinforcement Learning (RL) methods [7, 43] fail to solve these challenges. While rule-based methods lack adaptability to the dynamic online e-commerce environment [7], while RL-based methods rely on the Markov assumption, failing to capture sequential historical dependencies and temporal correlations [8, 13, 27].

The above methods are limited by their lack of flexibility and inability to capture value dependencies and temporal correlations. The rise of the generative paradigm, especially the successful application of Decision Transformer (DT) [8] in offline reinforcement learning, offers an innovative solution for complex sequential decision-making tasks. By reframing the decision process as a conditional sequence modeling task, DT effectively captures the context and temporal dependencies of historical marketing decisions.

However, directly deploying DT in the auto-marketing faces several critical challenges. First, limited by the quality and coverage

of the training dataset, DT may fail to learn the optimal strategy solely through offline training. Second, the standard DT paradigm conditions its policy on Return-To-Go (RTG) during training. However, using only reward-based RTG fails to express the objective of maximizing the marketing objective while ensuring constraint satisfaction [15, 16]. Third, existing research generally overlooks post-training alignment of DTs, resulting in reliance on periodic retraining of the foundation model to adapt to changing business needs, which leads to training overhead and resource waste.

To overcome these limitations, we propose a DT-based Generative Auto-Marketing framework (GAM), which provides an end-to-end generative solution for the auto-marketing problem, supporting the alignment of pretrained policy model to adapt to marketing objectives. First, we reframe the auto-marketing problem as a joint sequential decision-making task [34] and adopt DT to jointly optimize bidding and coupon discounting decisions. By learning a stochastic policy, we provide the policy model with effective exploration capability to mitigate out-of-distribution risk. Second, we propose a constraint-aware reward preference alignment paradigm, using reinforcement learning to optimize pretrained policy model in the post-training phase. Specifically, we first introduce a Marketing Dynamic Prediction Network (MDPN) for offline evaluation of the policy model. Next, to guide policy updates, we construct a Constraint-Aware Reward Model (CARM) to transform the self-funding constraint into a component of the reward signal. Finally, we utilize Group Relative Policy Optimization (GRPO) [32] for efficient fine-tuning, thereby avoiding the expensive overhead of repeatedly retraining the foundation model, which offers important practical significance.

In summary, our contributions are:

- We introduce an innovative marketing service Marketing Management in online e-commerce platforms, and are the first to study the auto-marketing problem in this scenario, providing a novel research perspective.
- We model the auto-marketing problem as a sequential decision-making problem and propose an end-to-end generative framework, GAM. GAM leverages DT to jointly optimize bidding and coupon discounting decisions, effectively addressing the limitations of commonly used methods. Furthermore, GAM employs reinforcement learning to align pretrained generative model with complex business objectives.
- We validate the effectiveness and adaptability of our proposed approach through extensive experiments.

2 Related Work

2.1 Auto-bidding and Coupon Discounting

Auto-bidding helps advertisers maximize their advertising objectives under constraints by adaptively optimizing bids for each impression [16, 41]. Early approaches primarily rely on predefined rules [9, 44] and online feedback [42, 45] to adjust bidding strategies. With the increasing complexity of online bidding environments, reinforcement learning has become crucial for auto-bidding. USCB [16] proposes a unified RL solution for the constrained auto-bidding problem. With the emergence of the generative paradigm, works such as Diffbid [15], CBD [26], GAVE [13], and GRAD [24]

utilize diffusion models and transformers to optimize bidding trajectories.

Coupon discounting is a core marketing strategy in e-commerce platforms for incentivizing consumption [5]. Prior approaches model this as an online knapsack problem [3, 14, 25] or utilize bandit feedback for strategy optimization [19, 28, 35, 36]. Current mainstream paradigms involve using machine learning [47, 49, 50] and reinforcement learning [38, 39, 46] for end-to-end optimization of coupon distribution strategies.

2.2 Offline Reinforcement Learning and Decision Transformer

Offline reinforcement learning focuses on learning effective policies from a static dataset without requiring online interaction, with methods like BCQ [12] and CQL [23] effectively solve the issue of value function overestimation, and IQL [22] reduces distribution shift through implicit policy improvement. However, RL-based methods rely on the Markov assumption, limiting their ability to model sequential dependencies. Generative models, represented by Decision Transformer (DT) [8], overcome this limitation by reframing offline RL as a conditional sequence modeling task. Methods like QT [17], QCS [20] enhance DT's ability to extract high-return policies from dataset, and ODT [48] introduces a sample-efficient online learning framework.

2.3 Preference Alignment

Preference alignment aims to fine-tune a pretrained generative model to specific preferences, thereby guiding the model to generate outputs that align with human preferences. RLHF [30] optimizes model generation quality by using human feedback as reward signal. DPO [31] and its extensions [4, 10, 29, 40] directly optimize the base model using human preference datasets. GRPO [32] improves the stability and efficiency of fine-tuning by adopting group-relative advantage estimation and reward functions. To the best of our knowledge, we are the first to explore reward preference alignment for DT-based generative models, enabling them to adapt to different preferences without costly retraining.

3 Preliminary

3.1 Auto-Marketing Problem Formulation

We consider the auto-marketing problem, where an auto-marketing agent automatically adjusts marketing decisions on behalf of merchant. In a certain period of time, assume that I impression opportunities arrive sequentially, indexed by $i \in [I]$. The auto-marketing agent participates in real-time competition by submitting a bid b_i and a coupon discount q_i for each opportunity, aiming to maximize the cumulative value of won impressions. The agent wins impression i if its bid b_i exceeds the highest bid from other participants. Upon winning, the merchant obtains a value v_i and the agent pays the bidding cost c_i and the coupon cost $q_i v_i$. Under a generalized second-price auction mechanism, bidding cost c_i equals the second-highest bid. The objective of the auto-marketing agent is to maximize the cumulative value of won ad impressions within this period:

$$\max \sum_i v_i x_i \quad (1)$$

where $x_i \in \{0, 1\}$ is a binary indicator denoting whether the agent wins impression i . The value v_i of impression i is calculated using metrics such as conversion rate and click-through rate.

The agent's coupon discount q_i affects v_i by influencing the user's purchase propensity towards the advertised product. Specifically, a higher q_i typically enhances the user's purchase propensity, thereby increasing click-through rate and conversion rate, ultimately leading to higher impression value.

Simultaneously, the auto-marketing agent must satisfy the self-funding constraint to control marketing expenditure. This constraint requires that the total cost incurred by the agent for bidding and coupon discounting must not exceed the total commission revenue obtained from the value of won impressions during that period. Compared to merchants manually setting static budgets and KPI constraints, the self-funding constraint alleviates the operational burden on merchants and provides the agent with a more flexible decision space. Specifically, for all I impression opportunities in this period, the agent's commission rate is fixed at α . For a won impression i , the agent pays the bidding cost c_i and the coupon cost $q_i v_i$, while receiving a commission revenue of αv_i . The self-funding constraint can be formalized as:

$$\sum_i c_i x_i + \sum_i q_i v_i x_i \leq \sum_i \alpha v_i x_i \quad (2)$$

Therefore, considering the marketing objective and the self-funding constraint, the entire process can be represented as:

$$\begin{aligned} \max_{b_i, q_i} \quad & \sum_i v_i x_i \\ \text{s.t.} \quad & \sum_i c_i x_i + \sum_i q_i v_i x_i \leq \sum_i \alpha v_i x_i \end{aligned} \quad (3)$$

3.2 Generative Auto-Marketing

The challenges in solving the auto-marketing problem primarily stem from the uncertainty of future marketing performance, and the complex coupling of the joint decision between bid b_i and coupon discount q_i . For this complex joint decision problem, a straightforward online learning solution introduces an independence assumption, decoupling the original problem into a two-stage optimization of coupon discounting and bidding. The intuition behind this decoupled paradigm is to set the coupon discount q at the initial step and keep it fixed in the period to determine the upper bound of the bidding constraint. This simplifies the bidding decision in the period to a sub-problem: maximize the cumulative value of won impressions under a KPI constraint, which yields an optimal uniform bidding coefficient λ^* . However, this two-stage paradigm, based on the decoupling assumption, cannot guarantee the global optimality of its solution because it fails to directly learn from the joint distribution of bidding and coupon discounting.

Moreover, for the highly dynamic online e-commerce platform, the optimal marketing decision should be adjusted iteratively to maximize the cumulative value [27]. This leads us to formulate the auto-marketing problem as a sequential decision-making task. Commonly used approaches for solving such tasks rely on rule-based strategies or RL techniques. However, rule-based strategies often fail to adapt to the high dynamism and uncertainty of the online e-commerce environment. Furthermore, the Markov property of RL

dictates that its state transitions typically focus only on information from the current timestep, making it difficult to capture complex sequential historical dependencies, such as the ad carryover effect [6].

To address this limitation, we adopt the DT paradigm for generative sequence modeling. The generative paradigm inherently possesses the ability to learn from high-dimensional joint distributions, which allows us to directly convert the original problem into a joint sequential decision-making task, without relying on the strong independence assumption used for decoupling. Ultimately, we transform the complex joint decision problem of auto-marketing into a sequential decision-making task for iteratively identifying the optimal marketing decision (λ^*, q^*) , thereby achieving end-to-end joint decision-making.

3.3 Sequential Modeling of Auto-Marketing

Under the sequential modeling of auto-marketing, we initially partition the marketing episode into T discrete timesteps. At each timestep t , the agent receives a real-time marketing state $s_t \in S$ and accordingly selects a marketing decision as the action $a_t \in A$. The agent's policy is expressed as $\pi(a_t|s_t)$. The state transition dynamics of the marketing environment are unknown, and the next state $s_{t+1} \in S$ depends not only on the current state and action but also on the historical trajectory τ . After transitioning to the next state, the marketing environment returns a reward r_t , which represents the cumulative value obtained during period t . The agent's objective is to maximize the total received value $R = \sum_t r_t$. A detailed description of our modeling is as follows:

- **State s_t :** The state s_t describes the marketing status at timestep t . In the auto-marketing scenario, the state can include the remaining time of marketing episode, historical statistics, cost consumption rate, and the current constraint satisfaction status, etc.
- **Action a_t :** The action a_t represents the marketing decision that is iteratively updated at each timestep t within the marketing episode. It is defined as the joint prediction of the uniform bidding parameter λ_t and the coupon discount q_t , i.e., $a_t = (\lambda_t, q_t)$.
- **Reward r_t :** Assuming $\mathcal{O}_t$ is the set of candidate impressions between timestep t and $t + 1$, the reward r_t is defined as the total value of won impressions in $\mathcal{O}_t$:

$$r_t = \sum_{i \in \mathcal{O}_t} v_i x_i \quad (4)$$

- **Return-To-Go (RTG) rtg_t :** The RTG value rtg_t represents the sum of rewards to be obtained in the future timesteps:

$$rtg_t = \sum_{t'=t}^T r_{t'} \quad (5)$$

where T is the final timestep.

4 Method

4.1 DT-based Generative Model

We adopt the DT architecture as the backbone network for generative decision-making in GAM, the overall framework is presented in Figure 1. DT generates the optimal policy by performing offline learning on historical trajectories τ , and it utilizes the

self-attention mechanism to effectively capture the necessary complex historical dependencies and temporal correlations during the decision-making process. DT processes the trajectory τ into an input sequence of RTG, states, and actions, denoted as (rtg_t, s_t, a_t) for timestep t . At timestep t , DT auto-regressively generates the action a_t using the trajectory representations of the most recent L timesteps; L is also referred to as the context length of DT. For subsequent descriptions, we use $(rtg_{t-L:t}, s_{t-L:t}, a_{t-L:t-1})$ as the abbreviation for the context sequence.

We note that if the model is trained directly using the standard RTG implementation shown in Equation (5), the resulting policy will completely prioritize maximizing the cumulative value, leading to severe violations of the self-funding constraint. To address this issue, we compute the Future Benefit-Cost Ratio (FBCR) at each timestep t , defined as:

$$\text{FBCR}_t = \frac{\sum_{t'=t}^T e_{t'}}{\sum_{t'=t}^T p_{t'}} \quad (6)$$

where e_t and p_t are the commission revenue and marketing cost during time period t , respectively. Assuming $\mathcal{O}_t$ is the set of candidate impressions between timestep t and $t + 1$, the commission revenue is $e_t = \sum_{i \in \mathcal{O}_t} \alpha v_i x_i$, and the marketing cost is $p_t = \sum_{i \in \mathcal{O}_t} (c_i + q_i v_i) x_i$. For a timestep t , $\text{FBCR}_t \geq 1$ indicates that the future trajectory satisfies the self-funding constraint; otherwise, the constraint is violated.

We ultimately use the FBCR to modify the standard RTG implementation, thereby integrating the constraint into the training stage:

$$\widetilde{rtg}_t = rtg_t + \lambda \text{FBCR}_t \quad (7)$$

where λ is a constraint weight hyperparameter.

Unlike the standard DT implementation, which learns a deterministic policy $\pi_\theta(a_t|rtg_{t-L:t}, s_{t-L:t}, a_{t-L:t-1})$, we opt to learn a stochastic policy that maximizes the dataset likelihood. This transformation of the learning objective is based on a natural observation: strategies trained directly on offline datasets are often sub-optimal due to the limitations in the quality and coverage of the training data. A straightforward approach to resolving this sub-optimality is to introduce action exploration during training to guide policy improvement. However, the objective of action exploration is fundamentally misaligned with the supervised learning objective of DT. This misalignment leads us to shift towards a post-training alignment paradigm, dividing the generative model's policy learning into two phases: offline pre-training and post-training alignment. However, during the alignment phase, the deterministic policy learned by standard DT struggles to sample high-return trajectories that may lie outside of distribution. Furthermore, it cannot directly apply widely used policy-based RL algorithms for finetuning. Therefore, we adapt DT to learn a stochastic policy, which inherently endows the policy model with an exploration capability, and enables the application of policy-based RL techniques for alignment during the post-training phase.

In the setup of GAM, for the two-dimensional continuous action space $a_t = (\lambda_t, q_t)$, which is composed of the uniform bidding parameter λ_t and the coupon discount q_t , we characterize the stochastic action policy based on state and RTG using a multivariate

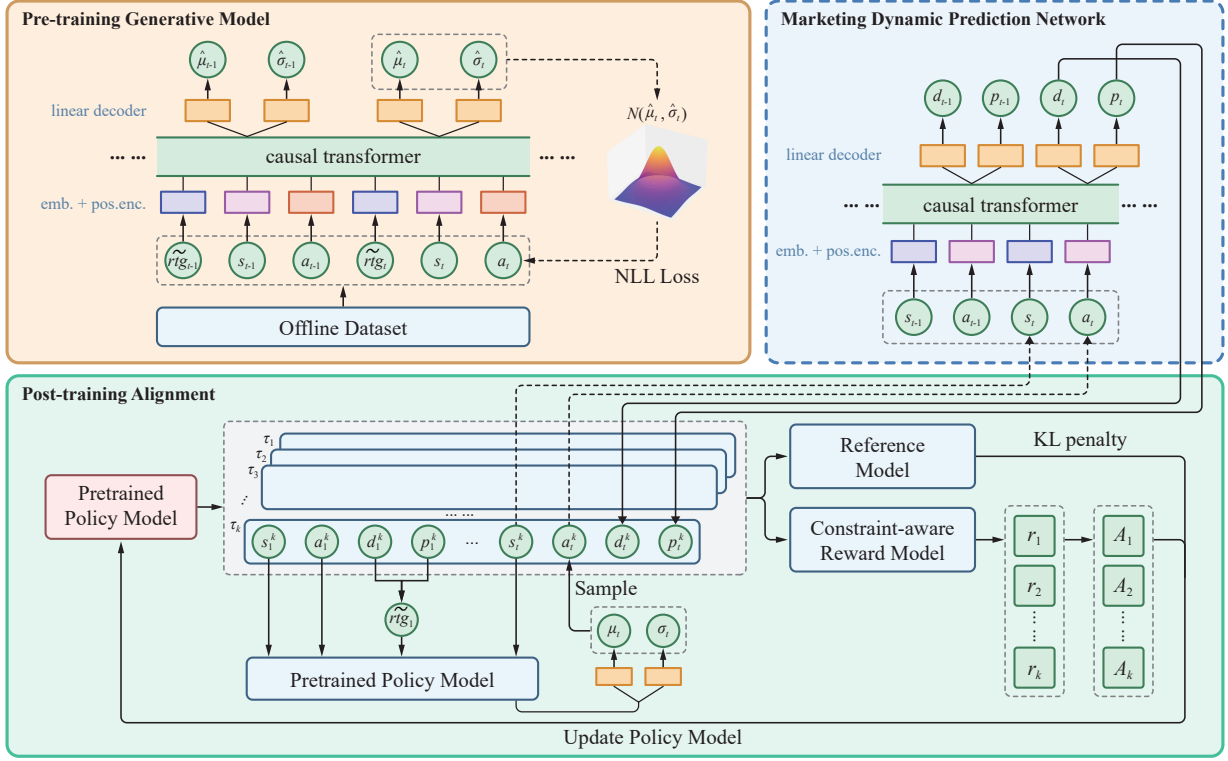


Figure 1: Overall framework of GAM

Gaussian distribution with a diagonal covariance matrix. Specifically, our stochastic policy is:

$$\pi_\theta(a_t | \tilde{r}g_{t-L:t}, s_{t-L:t}, a_{t-L:t-1}) = \mathcal{N}(\mu_\theta(a_t | \tilde{r}g_{t-L:t}, s_{t-L:t}, a_{t-L:t-1}), \Sigma_\theta(a_t | \tilde{r}g_{t-L:t}, s_{t-L:t}, a_{t-L:t-1})) \quad (8)$$

During training, to fit the historical data distribution, our objective is to maximize the log-likelihood of the stochastic policy over the trajectories in the training dataset. This is equivalent to minimizing the Negative Log-Likelihood (NLL) loss:

$$L_{NLL} = \frac{1}{L+1} \mathbb{E}_{(\tilde{r}g, s, a) \sim \tau} \left[- \sum_{t=L}^t \log \pi_\theta(a_t | \tilde{r}g_{t-L:t}, s_{t-L:t}, a_{t-L:t-1}) \right] \quad (9)$$

4.2 Post-training with Preference Alignment

Constrained by the bias of the training dataset, pre-trained generative model often falls into local optima. To resolve this issue, we employ the reward preference alignment paradigm during the post-training phase. In this section, we first introduce the Marketing Dynamic Prediction Network (MDPN) used for reward calculation, and then present the reward preference alignment module.

4.2.1 Marketing Dynamic Prediction Network. In the post-training alignment phase of GAM, we need to evaluate the potential efficacy of the policy π_θ in a real marketing environment. Since directly evaluating a new policy in a real online e-commerce platform is extremely costly and difficult to implement, we integrate a Marketing Dynamic Prediction Network (MDPN) into GAM. MDPN

is trained on historical trajectories and is used to estimate the impact of marketing decisions selected by the new policy on business metrics such as transaction volume and marketing cost.

MDPN utilizes a Transformer-based architecture to capture sequential dependencies within marketing dynamics and employs Zero-inflated Projection (ZIP) [18] to model the unique zero-inflated distribution of marketing data. Specifically, MDPN predicts the non-zero probability of target variables via a binary classifier and incorporates this estimated probability into the final value prediction, thereby achieving a joint optimization objective for both classification and regression. At each timestep t , MDPN takes the historical state-action trajectory $(s_{t-L:t}, a_{t-L:t})$ as input and predicts the transaction volume d_t and marketing cost p_t for time period t . The training objective is to minimize a joint loss function combining Cross-Entropy (CE) and Mean Squared Error (MSE) losses, defined as:

$$\mathcal{L}_{d_t} = \Pr(d_t \neq 0) \cdot \mathbb{I}_{\{d_t \neq 0\}} + (1 - \Pr(d_t \neq 0)) \cdot \mathbb{I}_{\{d_t = 0\}} + (d_t - \hat{d}_t)^2 \quad (10)$$

$$\mathcal{L}_{p_t} = \Pr(p_t \neq 0) \cdot \mathbb{I}_{\{p_t \neq 0\}} + (1 - \Pr(p_t \neq 0)) \cdot \mathbb{I}_{\{p_t = 0\}} + (p_t - \hat{p}_t)^2 \quad (11)$$

$$L_\phi = \frac{1}{L+1} \mathbb{E}_{(s,a) \sim \tau} \left[\sum_{t'=t-L}^t (\mathcal{L}_{d_{t'}} + \mathcal{L}_{p_{t'}}) \right] \quad (12)$$

MDPN serves as a simulation environment to roll out future trajectories. Specifically, given the historical trajectory $(s_{t-L:t}, a_{t-L:t-1})$ and the sampled action $a_t \sim \pi_\theta$, MDPN predicts the transaction volume d_t and marketing cost p_t for time period t . These predicted

values are then used to compute the state s_{t+1} for the next timestep, allowing us to collect the output trajectory timestep by timestep. The detailed roll out procedure is provided in Algorithm 1.

Algorithm 1: Trajectory Roll Out Procedure

```

1 Input: Policy model  $\pi_\theta$ ; MDPN; a batch of post-training
   dataset  $\mathcal{D}_b$ ;
2 Initialize trajectory buffer  $\mathcal{B} \leftarrow \emptyset$ ;
3 for initial parameter  $w \in \mathcal{D}_b$  do
4   Instruct the initial state  $s_1$  from  $w$ ;
5   for timestamp  $t = 1, \dots, T$  do
6     Sample action  $a_t \sim \pi_\theta(\cdot | s_{1:t}, a_{1:t-1})$  in an
       auto-regressive way;
7     Predict marketing dynamic
        $(d_t, p_t) \leftarrow \text{MDPN}(s_{1:t}, a_{1:t-1})$ ;
8     Instruct next state  $s_{t+1}$  from  $(s_{1:t}, a_{1:t-1}, d_{1:t}, p_{1:t})$ ;
9   end
10  Update trajectory buffer
      $\mathcal{B} \leftarrow \mathcal{B} \cup \{\tau := (s_t, a_t, d_t, p_t)_{t=1}^T\}$ ;
11 end
12 Output: trajectory buffer  $\mathcal{B}$ ;

```

4.2.2 Constraint-aware Reward Preference Alignment. The core challenge of auto-marketing is to maximize the marketing objective while avoiding deficits, which means integrating the constraint satisfaction objective into the reward preference alignment framework.

To address this, we construct the Constraint-Aware Reward Model (CARM). CARM uses a marketing objective function that includes a penalty term sensitive to the self-funding constraint as the reward signal. It further allows for adjusting the emphasis on constraint satisfaction through a constraint emphasis hyperparameter ρ . CARM incorporates the penalty term for violating the self-funding constraint to guide the model toward an optimal balance between marketing objective optimization and constraint satisfaction during the alignment phase:

$$\begin{cases} \text{BCR} = \frac{\sum_t e_t}{\sum_t p_t} \\ \text{P}(\text{BCR}) = \min\{(\text{BCR})^\rho, 1\} \\ r = \text{P}(\text{BCR}) \cdot \sum_t d_t \end{cases} \quad (13)$$

The term BCR represents the Benefit-Cost Ratio of the entire marketing trajectory, which differs from the Future Benefit-Cost Ratio (FBCR) computed in Equation (6). ρ is the constraint emphasis hyperparameter. In the actual roll out process, the trajectory-level BCR is cumulatively calculated based on the transaction volume d_t and marketing cost p_t predicted by MDPN at each timestep t using the current state s_t and action sample a_t .

Subsequently, we utilize GRPO [32] to optimize the pre-trained base policy model. GRPO transforms the constraint-aware reward signal into policy update gradients and ensures the stability of policy updates through group-relative advantage estimation. In more detail, for the initial parameter q of each marketing trajectory in

the post-training dataset, we perform K auto-regressive, timestep-by-timestep rollouts based on the old policy $\pi_{\theta_{\text{old}}}$ and MDPN, yielding a set of output trajectories $\{\tau_1, \tau_2, \dots, \tau_K\}$. For each trajectory τ_k in the set, we compute a trajectory-level reward r_k based on the standard choice. This reward is then normalized by subtracting the group mean and dividing by the group standard deviation, which gives the group-relative advantage $\hat{A}_{k,t}$ corresponding to each timestep t in the trajectory:

$$\hat{A}_{k,t} = \frac{r_k - \text{mean}\{r_1, \dots, r_K\}}{\text{std}(\{r_1, \dots, r_K\})} \quad (14)$$

We optimize the policy model by maximizing the following objective function. This objective combines the group-relative advantage estimation with a KL divergence penalty, ensuring efficient and stable policy updates:

$$\begin{aligned} \mathcal{J}_{\text{GRPO}}(\theta) = & \mathbb{E}_{q \sim P(Q), \{\tau_k\}_{k=1}^K \sim \pi_{\theta_{\text{old}}}(\tau|q)} \left\{ \frac{1}{K} \sum_{k=1}^K \frac{1}{|\tau_k|} \sum_{t=1}^{|\tau_k|} \left[\right. \right. \\ & \min \left(\frac{\pi_\theta(\tau_{k,t}|q, \tau_{k,<t})}{\pi_{\theta_{\text{old}}}(\tau_{k,t}|q, \tau_{k,<t})} \hat{A}_{k,t}, \text{clip} \left(\frac{\pi_\theta(\tau_{k,t}|q, \tau_{k,<t})}{\pi_{\theta_{\text{old}}}(\tau_{k,t}|q, \tau_{k,<t})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{k,t} \right) \\ & \left. \left. - \beta \mathbb{D}_{\text{KL}}[\pi_\theta \parallel \pi_{\text{ref}}] \right] \right\} \end{aligned} \quad (15)$$

ϵ and β are training hyperparameters.

By employing constraint-aware reward during the post-training alignment phase, GAM transforms the self-funding constraint—a necessity in the auto-marketing problem—into part of the optimization objective and integrates it into the policy update gradients.

Furthermore, GAM's base generative model outputs a randomized policy instead of a deterministic one, inherently possessing exploration capability in out-of-distribution (OOD) regions. This allows for the efficient sampling of high-return trajectories during the post-training alignment phase for policy updates. Consequently, GAM helps the policy model escape the local optima caused by the limitations of the pre-training data, achieving an empirical global balance between maximizing the business objective and satisfying the self-funding constraint.

5 Experiment

To evaluate the effectiveness of GAM, we construct a simulation environment based on real-world dataset and conduct extensive simulation experiments.

5.1 Experimental Setup

5.1.1 Simulation Environment. We construct the simulation environment using marketing logs from a major e-commerce platform. Specifically, we collect marketing logs for 25,000 products utilizing Marketing Management service over 50 consecutive days. We use 90% of data as the training dataset for MDPN, and the remaining 10% as the test dataset to provide initial parameters for each simulation episode, such as commission rate and product features. We treat the marketing log for a single product within one day as an episode. A complete episode is uniformly divided into $T = 24$ timesteps, with each timestep treated as the basic unit of decision-making.

In each episode of the simulation environment, the auto-marketing agent predicts the marketing decision a_t (i.e., the uniform bidding parameter λ_t and coupon discount q_t) at timestep t . Subsequently, MDPN evaluates the agent's policy based on the agent's decision and the historical state-action sequence of the current episode, returning an estimated transaction volume d_t and marketing cost p_t for time period t .

5.1.2 Evaluation Metrics. Based on the objective of auto-marketing, we use the following metrics to evaluate performance:

- **Average Return on Investment (ROI):** The ratio of the total commission revenue to the total marketing cost for the agent over the entire simulation period, defined as

$$\text{ROI} = \frac{\sum_m \sum_t e_t}{\sum_m \sum_t p_t} \quad (16)$$

- **Average Reward (AR):** The average transaction volume obtained by the agent in each episode over the entire simulation period, defined as:

$$\text{AR} = \frac{1}{M} \sum_m \sum_t d_t \quad (17)$$

- **Average Score (AS):** We use the score-based reward in Equation (13) to compute the average score obtained by the agent in each episode, where $\rho = 2$:

$$\text{AS} = \frac{1}{M} \sum_m \mathbb{P}(\text{BCR}_m) \sum_t d_t \quad (18)$$

5.1.3 Baselines. To validate the effectiveness of GAM in the simulation experiments, we compare it against a series of baselines:

- **Two-Stage Optimization (TSO):** TSO decouples the joint decision problem in auto-marketing into a two-stage optimization of coupon discounting and bidding.
- **IQL[22]:** IQL performs implicit policy evaluation by applying expectile regression, thereby avoiding distribution shift without requiring access to out-of-scope actions.
- **DT[8]:** DT employs the Transformer architecture for sequential decision-making modeling. In our implementation, we refer to the modification in Equation (7) to integrate the constraint into the pre-training stage.
- **GAVE[13]:** GAVE enhances DTs for offline generative auto-bidding through value-guided explorations.
- **GAS[27]:** GAS implements a DT-based offline bidding framework with post-training search by applying Monte Carlo Tree Search (MCTS)[21].

5.1.4 Implementation. We deploy the baseline models following the standard implementation, with necessary modifications made to suit the experiment requirements, such as extending the action space to accommodate joint decision-making. All experiments are conducted on NVIDIA A100 GPUs using the PyTorch framework, with the batch size set to 256. GAM's base policy model follows the standard DT implementation, adopting a Causal Transformer architecture with 8 attention layers and 16 attention heads, and setting the hidden layer size to 512. During training, we optimize the model parameters using the AdamW optimizer with a learning rate of 1e-5. The MDPN within GAM shares a largely identical architecture and parameter setting with the base policy model.

However, MDPN generates deterministic outputs instead of a stochastic distribution and does not use RTG as input during training. We implemented the GRPO algorithm to be compatible with the DT architecture by referencing the original paper [32] and a public implementation [37], and we set the hyperparameters based on default values. Unlike standard GRPO practice, GAM's rollout procedure requires the next state at the current timestep to depend on the model's action output and the simulation environment's feedback. This necessitated modifying the standard practice's output generation and policy probability calculation components.

5.2 Experimental Results

We conducted a comprehensive comparison of GAM's performance against various baseline methods in the simulation experiments, demonstrating the effectiveness of GAM over the baselines. The experimental results are summarized in Table 1.

Table 1: Comparison of GAM with baselines in the simulation experiment. The best performance is highlighted in bold.

Model	ROI	AS	AR
TSO	82.77%	18.70	20.77
IQL	72.05%	9.96	19.83
DT	70.57%	14.78	19.41
GAVE	88.63%	15.98	19.82
GAS	92.45%	16.66	20.81
GAM	109.16%	22.22	23.63

Our results show that the rule-based TSO method exhibits competitive performance due to its intuitive heuristic design. However, it is limited by policy adjustment latency caused by decoupling bidding and coupon discounting decisions, making it difficult to reach global optimality. Furthermore, DT-based methods (DT, GAVE, and GAS) significantly outperform the traditional RL method (IQL), further validating the effectiveness of generative architectures in solving complex sequential decision-making problems. Notably, GAM achieves superior performance across all metrics compared to advanced baselines like GAVE and GAS. This confirms that by utilizing joint decision optimization and combining the pre-trained generative model with post-training alignment techniques, GAM effectively overcomes the limitations of existing methods and successfully aligns with complex business objectives.

5.3 Alignment Analysis

One of the core challenges in auto-marketing is achieving a balance between optimizing business objectives and satisfying self-funding constraint, which necessitates adjusting the policy model's optimization objective according to business scenarios. GAM addresses this by employing a flexible reward model CARM, which allows the policy model to be aligned with various optimization objectives during the post-training phase.

We investigate GAM’s performance under different optimization objectives and evaluation metrics to verify its adaptability and flexibility to various business scenarios. Specifically, we consider three optimization objectives by modifying the reward function settings in Equation (13), defined as follows:

$$\begin{cases} r_1 = \sum_t d_t \\ r_2 = \min\{(BCR)^2, 1\} \cdot \sum_t d_t \\ r_3 = \min\{(BCR)^5, 1\} \cdot \sum_t d_t \end{cases} \quad (19)$$

Similarly, we modify the Average Score (AS) evaluation metric using the same modification, yielding the corresponding evaluation metrics AS_1 , AS_2 , and AS_3 . AS_1 only considers the optimization of the business objective, representing business scenarios with no constraint restriction. AS_2 is the default implementation in GAM, applying an appropriate penalty for constraint violation, representing the standard business scenario. AS_3 further increases the penalty coefficient for constraint violation, representing business scenarios with strict restrictions on self-funding constraint satisfaction.

Table 2: Alignment performance with different preferences.

Train\Eval	AS_1	AS_2	AS_3
r_1	24.54	10.61	6.72
r_2	23.63	22.22	20.04
r_3	22.06	21.00	20.13

The results, presented in Table 2, show that GAM achieves high-performance when its training objective is precisely aligned with the evaluation metric. This finding highlights the importance of aligning the post-training objective with business preferences to achieve optimal performance, and further demonstrates GAM’s ability to flexibly adapt to various optimization objectives through the incorporation of CARM.

5.4 Ablation Study

To evaluate the effectiveness of each component within GAM, we conducted an ablation study by evaluating the following modified versions of GAM:

- **GAM w/o Post-training:** To evaluate the contribution of the post-training alignment step to GAM’s performance, we remove the post-training module from GAM and only run the pre-trained policy model to predict marketing decisions.
- **GAM w/o MDPN:** To assess the necessity of constructing an offline policy evaluation environment, we remove MDPN from GAM and replace it with a rule-based predictor. This predictor does not rely on the model’s actual action-state trajectory; instead, it directly uses the average transaction volume $\bar{d}_t$ and average marketing cost $\bar{p}_t$ up to timestep t in the test dataset as the estimation of marketing dynamics.
- **GAM w/o CARM:** To evaluate the effectiveness of introducing the constraint signal into the reward preference alignment,

we remove CARM from GAM and only use the marketing objective as the reward signal during the post-training phase, i.e., $r = \sum_t d_t$.

Table 3: Ablation Study

Model	ROI	AS	AR
GAM w/o Post-training	72.20%	11.52	21.13
GAM w/o MDPN	96.08%	15.70	20.23
GAM w/o CARM	75.49%	10.61	24.54
GAM	109.16%	21.69	23.14

The results in Table 3 demonstrate that all core modules of GAM are indispensable for achieving optimal performance. Specifically, GAM w/o Post-training significantly underperforms GAM, confirming the importance of preference alignment in post-training. Simultaneously, GAM w/o CARM yields inflated AR at the expense of a precipitous ROI and AS decline, this indicates that CARM’s translation of the self-funding constraint into a constraint-aware reward is decisive for guiding policy updates toward constraint-compliant regions. Furthermore, the performance degradation of GAM w/o MDPN validates MDPN’s indispensability as a sequence-aware dynamic prediction network for providing high-quality, action-dependent rollout evaluation for post-training.

6 Conclusion

This paper introduces Marketing Management, an innovative marketing service in e-commerce platforms, and is the first to study the auto-marketing problem in this scenario, addressing the limitations of classic auto-bidding. We model the auto-marketing problem as a sequential decision-making problem and propose a novel generative auto-marketing framework, GAM, addressing the dual challenges in auto-marketing: the self-funding constraint and the joint decision of bidding and coupon discounting. GAM utilizes DT architecture to reframe the joint decision-making as conditional auto-regressive sequential modeling, thereby effectively capturing the coupled dependencies between bidding and coupon discounting. Furthermore, by integrating a constraint-aware preference alignment module into GAM, we guide the model to achieve a balance between maximizing the business objective and constraint sanctification. Through extensive experiments, we demonstrate the effectiveness and adaptability of GAM, providing a general solution for automatically optimizing marketing activities on e-commerce platforms.

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